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RESEARCH ARTICLE

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Assessment of spawning habitat suitability for Amphidromous Ayu (*Plecoglossus altivelis*) in tidal Asahi River sections in Japan: Implications for conservation and restoration

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Abstract

The Amphidromous Ayu (*Plecoglossus altivelis*), vital for Japan's riverine commercial fisheries, is experiencing population decline due to habitat degradation. In response, Japan is implementing restoration efforts by releasing juvenile fish and cultivating spawning habitats. This study, focusing on the Asahi River's Heidan area, evaluates Ayu spawning habitat suitability, especially in tidal zones, and its conservation implications. Through field surveys and numerical analysis, the research identifies optimal conditions for spawning amid dwindling numbers. Observations from the 2020 peak spawning season showed that water temperatures below 20°C are conducive to spawning, with the best sites producing up to 48,817 eggs/m^2. Gravel sizes between 0.85 and 53.0 mm were identified as crucial for effective spawning. Notably, artificial spawning grounds established in 2019 showed no spawning activity, likely due to saline intrusion from tides. Using the three-dimensional hydrodynamic-aquatic ecosystem numerical model, the study assesses flow, substrate, and water quality's impact on spawning, corroborated by sensor data. A key finding is the negative correlation between salinity and egg density in tidal sections of the Asahi River. Furthermore, an artificial intelligence (AI)-based YOLOv5 model, trained on underwater images, effectively detected Ayu aggregations, demonstrating an *F*1-score of 0.757 in differentiating Ayu from other coexisting fish species in the same aquatic environment. This AI approach provides a nonintrusive method for monitoring Ayu populations and habitat preferences. The research underlines that successful spawning habitats require minimal saline intrusion, with optimal salinity and flow conditions. These insights are critical for spawning ground development, emphasizing the management of tidal effects and riverbed conditions to bolster Ayu populations and preserve aquatic ecosystems.

KEYWORDS
artificial intelligence, Ayu spawning, habitat suitability, hydrodynamic-ecological modeling, saline intrusion

1 | INTRODUCTION

The Amphidromous Ayu (*Plecoglossus altivelis*), ranging from 10 to 30 cm, engages in annual migratory journeys between the sea and rivers, predominantly inhabiting Japan, Korea, and eastern China (Nagaya et al., 2008). This fish spawns in riverbeds with loose gravel, similar to Salmon and Trout, with eggs hatching within approximately 2 weeks at water temperatures around 15°C. The larvae migrate to the sea at dusk, subsisting on zooplankton for 3 to 5 months before returning to the river to feed on algae. By autumn, they migrate downstream to spawn when water temperatures fall below 20°C, typically between October and December in Japan (Takahashi & Azuma, 2006), after which most Ayu die. Figure S1 illustrates the life cycle of Ayu.

The Ayu, a key species in Japanese river fisheries with significant economic value in recreational fishing (Takahashi & Azuma, 2016), has seen a decline in catches in recent years, particularly in western Japan (Takahashi, 2009). In response, efforts are underway to bolster Ayu populations. These include modifying or eliminating hydraulic structures that impede the fish's movement, creating passage aids like stone piles, enhancing spawning areas, safeguarding breeding fish during spawning season, and releasing juveniles bred for resistance to 'cold water disease'. Additionally, the diminishing and aging workforce in inland fisheries highlights the urgency to refine spawning habitat improvement techniques and disseminate this expertise more widely. Our research aims to restore the Ayu's spawning habitats, correlating their degradation with population declines (Takahashi & Azuma, 2016). Identifying optimal spawning locations, considering water flow, water quality, and bed condition is crucial for Ayu preservation. Our study is centered on the Heidan region of the Asahi River in Okayama Prefecture, western Japan—a notable Ayu spawning hub. Given the declining Ayu catch in Okayama, it is crucial to pinpoint suitable spawning sites and enhance the spawning riverbed environment. Consequently, the Okayama Prefectural Agriculture, Forestry and Fisheries Research Center and the Nanbu Fisheries Cooperative Association have initiated artificial spawning ground projects in the Asahi River. A man-made spawning ground was established in Heidan in October 2019, with expansions in 2020 around natural spawning sites. However, the artificial sites experienced no spawning, prompting investigations into saline intrusion from tides affecting habitat selection.

Earlier studies emphasize the need to understand natural river ecosystem recovery to improve fish habitats (Choi et al., 2018; Konrad, 2009; Lee et al., 2010; Nagaya et al., 2008; Stoffers et al., 2022), particularly the relationship between physical and hydraulic habitats for disturbed river restoration. Despite extensive research on various fish spawning environments (Xing et al., 2021; Yang et al., 2023; Zhu et al., 2020), detailed studies on Ayu habitat preferences are limited. Over recent decades, particularly in Japan, efforts have been made to pinpoint ideal conditions for Ayu spawning in both freshwater and estuarine locations, examining factors like riverbed materials, flow patterns, and water quality. For instance, Ishida and Ichijo (1990) found a preference for small gravels in rapid zones with soft beds and loose stones. Onitsuka et al. (2005) linked spawning density to water depth and flow velocity through regression analysis and field observations. Building upon this, Nagaya et al. (2008) predicted optimal Ayu spawning flow conditions two-dimensional (2D) simulations and Physical HABitat SIMulation (PHABSIM) modeling, though model reliability decreased at higher flows. Fukui et al. (2016) refined PHABSIM spawning suitability assessments for Ayu, updating preference curves and considering sediment types. Furthermore, Yoshida et al. (2018) assessed shallow water spawning locations in the Asahi River using PHABSIM and high-resolution bathymetry using the airborne lidar topo-bathymetry (ALB) technique, providing insights into sediment dynamics and flow.

Drawing on earlier studies on substrate and hydraulic conditions, the Asahi River's Heidan region was chosen for restoration in 2019 to aid Ayu fish population recovery. However, no spawning was detected there, which was a significant observation given the correlation between tidal-induced saline water intrusion and Ayu spawning hindrance. Research by Iguchi et al. (2010) explored the effects of salinity on Ayu hatchlings, revealing that higher salinity and temperature quicken nutrient depletion, impacting early development and survival. Furthermore, Matsuda and Morita (2021) noted that hydraulic structures create stagnant zones that obstruct Ayu juvenile migration, leading to starvation. Thus, establishing spawning habitats in lower river areas, including tidal regions, is considered an effective approach for Ayu conservation. Despite the known relationship between river dynamics and Ayu spawning in fresh water, the influence of temporary saline water intrusion on spawning sites is less understood. Addressing this gap, we conducted a comprehensive survey of potential spawning beds in the lower Asahi River in Okayama Prefecture, Japan. This region is noteworthy because, to the best of our knowledge, it is one of the few instances in Japan where artificial spawning grounds have been established in tidal areas to help rejuvenate the Ayu population. Nevertheless, there has been a shortage of continuous measurements for water level and salinity, and the availability of flow data has been restricted. This lack of detailed spatial flow data in the complex Heidan topography, sensitive to tides, prompted us to utilize numerical analysis to elucidate temporal flow and saline patterns.

Moreover, to enhance our understanding of the Ayu fish's spawning environment in tidal river sections, it is essential to corroborate their migratory patterns, especially regarding optimal spawning habitats in artificially developed areas. The observation of Ayu fish poses several challenges. For the tidal sections of the Asahi River, precise identification techniques are essential to assess habitat suitability. Traditional methods involving fish capture, which rely on morphological characteristics such as coloration, fin structures, and body measurements (Teletchea, 2009), can be detrimental to the conservation of species and typically necessitate expert analysis, potentially leading to delays and inaccuracies in identification process. Alternatively, underwater cameras provide a noninvasive means to observe Ayu in their natural environment. Accurate identification relies on high-quality imagery to discern visual markers like shape, size, and behavior, which are crucial as per Baker et al. (2021). However, camera-based approaches can be compromised by environmental conditions such as turbidity and lighting. Also, it is worth noting that visual confirmation by human eyes requires considerable time and effort. The emergence of artificial intelligence (AI), and specifically deep learning, offers promising solutions for fish species recognition. AI-based models, like the You Only Look Once version 5 (YOLOv5) deep convolutional neural network, have demonstrated remarkable identification capabilities when trained with ample labeled images of fish species (Li et al., 2023; Liu et al., 2023), providing quick and consistent image analysis. Nonetheless, the effectiveness of these models is contingent upon a diverse and comprehensive training dataset to avoid inaccurate identifications. Integrating underwater cameras with AI appears to be the most effective strategy for monitoring Ayu populations in the Asahi River, which is instrumental for conservation and restoration efforts.

The overarching objective of this study is to determine the optimal spawning conditions (i.e., flow, riverbed composition, and water quality) for Ayu in tidal rivers, intending to facilitate the effective creation of spawning grounds through field observations and numerical simulations with the three-dimensional Hydrodynamic-Aquatic Ecosystem Model (AEM3D). Furthermore, the study seeks to confirm the migratory behavior of parent Ayu around these habitats by employing efficient observational techniques, including underwater cameras and AI-based modeling (YOLOv5) The outcomes of this research will inform strategies to replenish Ayu populations within the natural capacities of the rivers, thereby enhancing the overall health of the riverine ecosystem and benefiting both the natural environments and the human communities reliant on the river's vitality.

2 | STUDY SITE AND METHODOLOGY

2.1 | Target section and recovering project of Ayu

The Asahi River, the focal point of this study, is classified as a first-grade river in Japan, boasting a channel length of 142 km and a basin area of approximately 1810 km^2 (Yoshida et al., 2022). It meanders through the core of Okayama Prefecture before draining into the Seto Inland Sea, as illustrated in Figure 1a. The river's lower reaches are significantly influenced by the pronounced ebb-tide water level difference in Kojima Bay, its estuarine area. The flow dynamics within the Heidan region are dictated by both the tidal fluctuations of Kojima Bay (Figure 1b) and the river (Figure 1c). The Asahikawa Dam, positioned approximately 40 KP upstream, primarily governs these dynamics. The term 'kilo post' (KP) refers to the longitudinal distance in kilometers from the river's mouth. It is widely recognized that constructing upstream dams reduces riverbed disturbance during floods and diminishes the flow rate under normal conditions. The presence of an upstream dam might also indirectly affect the spawning habitat of Ayu fish and the migration of their larvae downstream by altering physical conditions such as flow velocity and saltwater intrusion. However, it remains unclear whether the Asahikawa Dam has a significant quantitative impact on the entire life cycle of Ayu. Additionally, the Asahikawa Dam lacks a fish passage; therefore, Ayu migrating upstream from the river mouth can only reach the area in front of the dam. Several potential factors could account for the decline in Ayu populations, one possibility being that restricted river channel capacity might impede their growth. However, there is no evidence to support this hypothesis in the specified river. Also, the water level fluctuation during high tide in the study site can reach approximately 2 m. Notably, around the 8.5 KP mark, the river meanders near a locally known site 'Korakuen' (Figure 1d), characterized by a deeply dredged channel where saltwater tends to remain stagnant.

Figure S2 depicts the decline in Ayu fish catches across Japan, with a notable decrease in Okayama Prefecture, raising concerns among fisheries officials. The data, provided by the Japanese Ministry of Agriculture, Forestry and Fisheries for the years 2003–2016, continues to be significant for current conservation efforts. In response, a restoration initiative was launched in Okayama Prefecture to create an artificial riverbed conducive to Ayu spawning. The primary objective of this undertaking between 2019 and 2020 was to enhance Ayu population recovery methods in the Asahi River. During the restoration phase, interventions were made to modify the riverbed environment: obstructive fine sand and mud were cleared, and bulky gravels, which pose a challenge for Ayu spawning, were extracted. Figure 1d shows the specific sites where these interventions were implemented to expand the Ayu spawning habitat. The focus on these spawning grounds in the river's tidal sections is due to the biological traits of the Ayu and the characteristics of Japanese rivers. Ayu larvae, after hatching, store nutrients for about 3 to 4 days before migrating to the ocean, which is rich in plankton. However, the flow in many Japanese rivers is often blocked by intake weirs, leading to stagnant water areas where newborn Ayu larvae are trapped, preventing them from reaching the ocean and resulting in starvation. To mitigate this, spawning grounds were strategically established in the tidal zone, facilitating a higher survival rate of Ayu larvae as they journey to the sea. In the Asahi River, additional potential spawning grounds were observed in freshwater areas beyond these tidal sections, influenced primarily by factors like flow velocity and depth. Nonetheless, the selection of optimal spawning sites is limited by various constraints, including unfavorable riverbed conditions.

2.2 | Field investigations

Field surveys were conducted during low tide due to significant tidal variations affecting water depth and rendering high-tide surveys impractical. Our investigations encompassed several parameters: spawning conditions (egg density/m^2), riverbed characteristics (softness and composition), water level and quality (salinity, temperature) monitored by sensors. Figure 1d displays the locations surveyed (P1, P2, P3, and P4), while Table S1 provides detailed information on the sensor-based water level and quality assessments. The comprehensive examination of spawning conditions, riverbed attributes, and sensor-monitored water quality measurements is elaborated in Supporting Information S1. Especially during egg density surveys, we added two reference sites located outside of the two artificial spawning grounds in the P1 and P2 areas, as shown in Figure 1d. These control areas play a critical role in providing baseline data on natural egg densities and environmental conditions. Figure 1d also represents the numerical estimation sites for hydraulic conditions, specifically water depth and flow velocity, denoted as P1 (a)–(c), P2 (a)–(c), and P3. Figure S5 depicts the detailed location of the sites P1 (a)–(c) and P2 (a)–(c). Additionally,

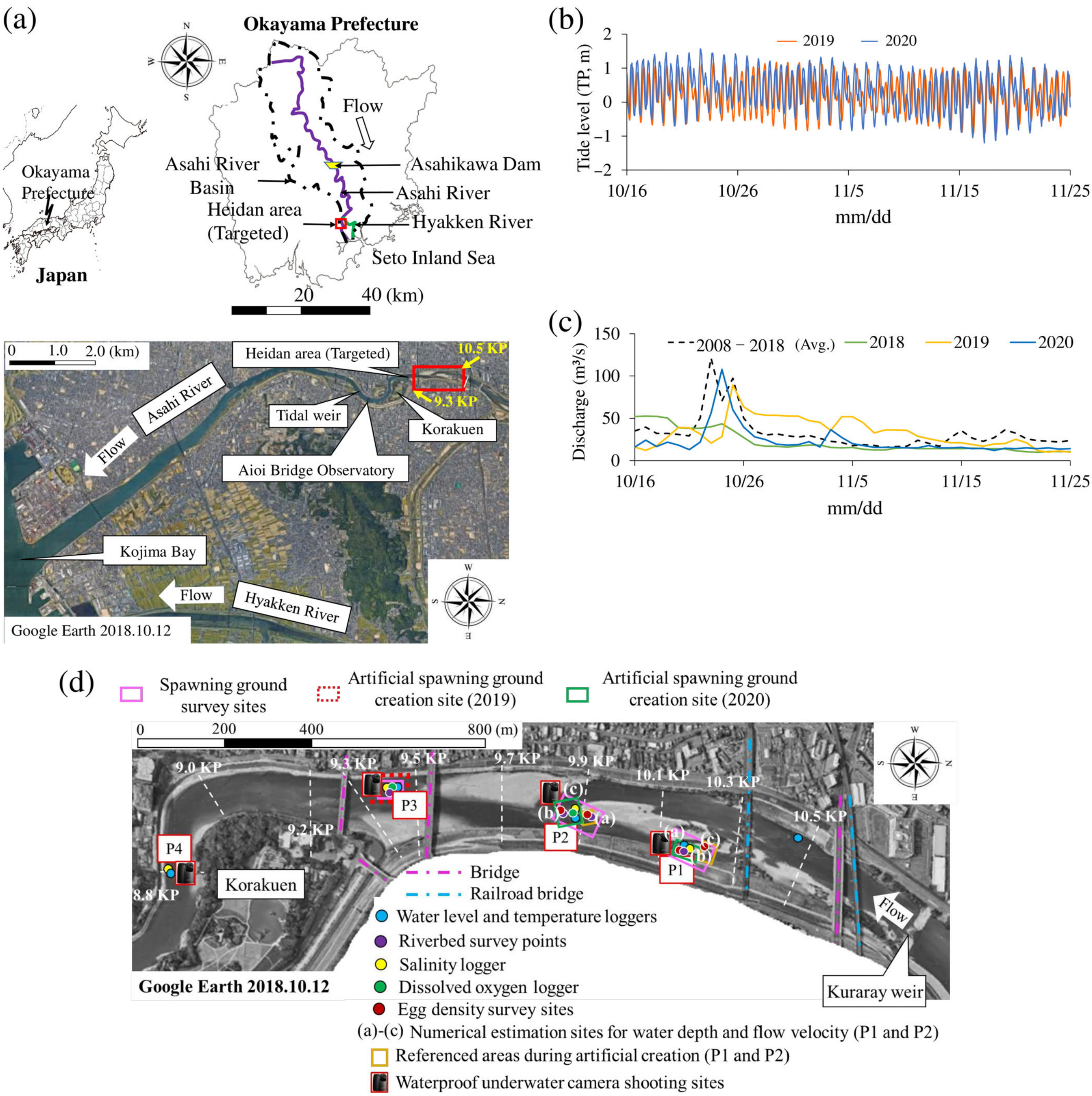

FIGURE 1 (a) Top: location of the Asahi River in Okayama Prefecture, Japan; bottom: Overview of the river's lower reaches spanning 0–12 KP (where KP, kilo post, indicates longitudinal distance in kilometers from the river mouth), (b) tide level at the mouth of the Asahi River in Kojima Bay (TP, Tokyo Peil as datum level representing the mean sea level of Tokyo Bay in Japan), (c) daily average discharge in the Heidan area of the lower Asahi River, and (d) targeted tidal areas featuring artificial spawning grounds accompanied by field observations. [Color figure can be viewed at wileyonlinelibrary.com]

waterproof underwater cameras were installed at the four targeted sites (P1–P4), as depicted in Figure 1d. These cameras captured images of various fish species to aid in the AI-based identification of Ayu fish migration patterns. Further details on underwater imaging and AI-based fish species detection are provided in Section 2.4. It is worth noting that Figure 1d does not provide a detailed depiction of the sensor and survey point locations.

2.3 | Numerical analysis

2.3.1 | Numerical model

We utilized AEM3D, a detailed 3D aquatic ecosystem model crafted by HydroNumerics Pty Ltd., Australia. This model integrates the estuary and lake computer model (ELCOM) from Hodges (2000) and

Oveisy et al. (2012) with the computational aquatic ecosystem dynamics model (CAEDYM) by Hipsey et al. (2006). We specifically explored the model's hydrodynamic module, as outlined by Zamani and Koch (2020) and Ulańczyk et al. (2021), to study how stratified water masses change over time due to different environmental forces. This module includes essential equations like the momentum, continuity, free surface (kinematic conditions), and conservation equations for mass, heat, and salinity, as detailed by Zamani and Koch (2020). Our methodology incorporated the unsteady 3D Reynolds-averaged Navier–Stokes equations to represent fluid motion and used the Boussinesq and hydrostatic approximations to define fluid properties.

2.3.2 | Datasets for numerical simulations

Bed elevation

Bed elevation data for the Asahi River (0.0–17.4 KP) was obtained through ALB measurements conducted on 5 February 2019. Table S2 provides the key specifications of the ALB device along with the measurement conditions applied in this study. Our approach to processing the ALB data involved the creation of a 2 m wide horizontal 2D cell containing 3D point clouds, following the methodology outlined by Yoshida et al. (2020). Through a selective filtering process targeting laser points near the lowermost portion of the 2D cell, we were able to determine the elevation of the ground or riverbed, referred to as the digital terrain model (DTM). A significant finding highlighted by Yoshida et al. (2020) was that ALB-derived DTM values could accurately replicate the riverbed profile, with an error range ranging from a few centimeters to several tens of centimeters. Consequently, the ALB data used in this study appears to hold promise and can be considered reliable for simulation purposes. Additionally, we utilized navigational data sourced from the Ministry of Land, Infrastructure, Transport, and Tourism. This data covers a numerical mesh size of approximately 50 $\times$ 50 m in the vicinity of the Takashima tidal area, gradually expanding to 200 $\times$ 200 m with increasing distance for both Kojima Bay and Yoshii River.

Initial and boundary conditions

The initial and boundary conditions in the analyzed section were set according to the details presented in Figure 2. The upstream flow of the Asahi River was sourced from the Makiyama hydraulic observatory (20 KP), utilizing the stage–discharge relationship, while the value for the Yoshii River was estimated at the Miyasu (14.05 KP) observatory. Throughout the calculation period, the Asahi River exhibited a peak runoff of 225 m^3/s on 23 October and a minimal runoff of 40 m^3/s on 4 November 2020. On all other days, the daily average flow ranged from 12 to 24 m^3/s (see Figure S3a). Regarding the relationship between runoff and spawning behavior, our earlier field surveys indicated that, even under normal flow conditions, some eggs were swept away from their spawning sites. This suggests that Ayu spawning eggs might be displaced along with riverbed sediment during high runoff, a phenomenon observed in other Japanese rivers. Furthermore, it is well-documented that the autumn runoff facilitates the downstream movement of adult Ayu to spawning sites and enhances their spawning activities (Takahashi, 2009). However, due to limited research, the impact of runoff on the net change in egg deposition remains unclear. Fisheries officials believe that during high runoff, adult Ayu likely move to the pools near Korakuen for refuge, returning

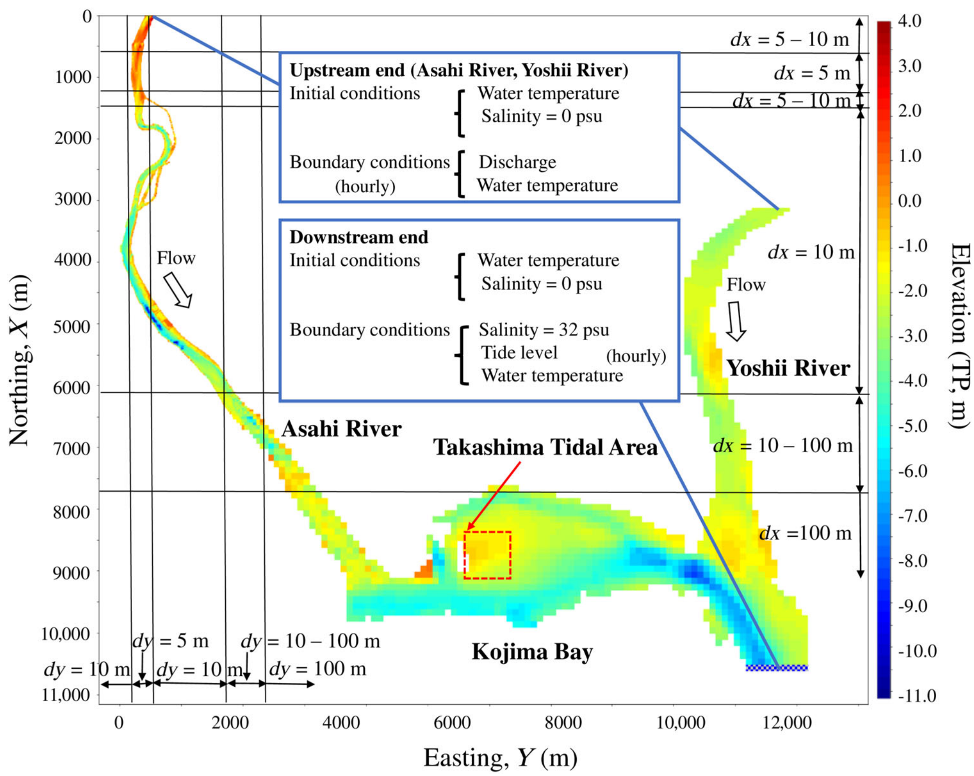

FIGURE 2 Numerical analysis domain with specified initial and boundary conditions. [Color figure can be viewed at wileyonlinelibrary.com]

to the riffles for spawning once river conditions stabilize. Therefore, our research extends beyond counting eggs to evaluating the spawning habitat by conducting field surveys and using underwater cameras to monitor adult Ayu aggregations around the spawning sites.

Furthermore, the downstream water level was estimated based on the hourly tide level at Uno Port, Tamano City, located in the Seto Inland Sea. It should be noted that the Uno Port is approximately 13 km away from the downstream end boundary considered in this study for numerical simulation. We performed a tentative numerical simulation using the estimated downstream water level data and confirmed the reproducibility of the observed water level at the mouth of the Asahi River. Adjustments were made to ensure a close alignment between these two water levels. Ultimately, the downstream water level was determined to be 1 h ahead and 15 cm higher than the tide level at Uno Port (see Figure S3b).

For the parameter of salinity, a constant value of 0 psu was assigned at the upstream end of both rivers, whereas a value of 32 psu was designated at the downstream end. It is worth noting that the salinity near the downstream end fluctuates due to the influence of tides and river currents. Therefore, in this study, we also took into account the salinity values near the downstream end, which were approximated based on records from previous years. The Asahi River's inflow water temperature was computed using data collected upstream of the Asahi River Bridge on the Japan Railway Sanyo Main Line in the Heidan area (see Figure S3c). Similarly, the inflow water temperature for the Yoshii River was determined using data obtained from the surface layer of Kuban near the river's mouth (see Figure S3c). Furthermore, at the downstream end, a uniform water temperature was assumed for the bottom layer at Kuban.

The meteorological boundary conditions incorporated in this study encompassed hourly data for wind direction, wind speed, pressure, temperature, humidity, rainfall, cloud cover, and shortwave radiation. These meteorological parameters were calculated at the Okayama Prefecture Meteorological Observatory, with the exception of shortwave radiation, which was derived from Himawari satellite data.

2.3.3 | Numerical conditions

In conducting numerical computations for the tidal section of the Asahi River, we defined the analysis domain to encompass the Heidan area and its downstream section (0–10.8 KP), Kojima Bay, and the downstream region of the Yoshii River (0–5.6 KP). To address the complex topography around the P1 point, where finer resolution was necessary, we initially utilized a 5×5 m horizontal grid spacing. Subsequently, we expanded the mesh spacing in the X-axis direction by 10%, resulting in a grid size of 10×10 m at the 4.0 KP location along the Asahi River and 100×100 m at Kojima Bay (refer to Figure 2). Furthermore, we adjusted the mesh spacing vertically, ranging from 0.1 to 0.2 m for areas above -6 m elevation and from 0.2 to 1.0 m for areas below -6 m elevation, with the spacing increasing proportionally with water depth. In summary, the total number of grid cells computed for the north-south, east-west, and vertical directions amounted to 739, 313, and 69, respectively. The calculation period spanned 43 days, from 11 October to 23 November 2020, incorporating a 4-day run-up period. During this time frame, we employed a time step of 5 s for the calculations.

2.4 | Using underwater cameras and AI to confirm Ayu fish migration

2.4.1 | Underwater camera shooting

We utilized the Brinno TLC200Pro time-lapse camera, a preferred tool in both scientific research and construction for recording extended footage (Vlaswinkel et al., 2023). We created a tailored waterproof casing to fit the camera for observing fish in the artificial habitat. The camera operated continuously on cell batteries. Images were captured every 10 s, resulting in 3-h daily recordings from 14:00 to 17:00 in ideal lighting, considering that spawning commonly takes place in the evening. All recordings from 4 October to 14 November 2020 were saved in AVI format with a resolution of 1280×720 pixels. To conduct our analysis, we needed to review approximately 360 images per hour, totaling 1080 images each day. The time spent reviewing images for a single day varied between 3 to 24 min, depending on the number of Ayu fish captured by the underwater camera. For a comprehensive analysis, we viewed the underwater footage on a computer and meticulously counted the Ayu fish.

2.4.2 | Image dataset creation for AI-based fish species detection

We created the Asahi Heidan Fish Dataset (AHF-D) by gathering images from diverse locations, specifically P1, P2, P3, and P4. This dataset encompasses a total of 3427 images, which are systematically categorized in Table S3, with a distribution of 2747 images for training, 338 for validation, and 342 for testing. These images were exclusively acquired during the training phase (Figure 8) of the comprehensive Ayu population survey, conducted between 4 October and 14 November 2020. To structure the AHF-D dataset, we divided it into three segments: training data, validation data, and test data, maintaining an 8:1:1 ratio for the targeted fish species, as outlined in Table 1. Specifically, we segmented the image dataset for AI-based detection, as detailed in Tables S3 and 1. In the artificial spawning grounds of the Asahi Heidan area, numerous fish species, aside from Ayu, are commonly found. To address this, we created a distinct test dataset, separate from the training data, and employed an AI-based model (i.e., YOLOv5) trained on the combined training and validation data to identify fish species within the test dataset. This allowed us to assess the model's ability to distinguish between different fish species. During the training phase, we ran 500 epochs, as it was the point at which the validation loss reached its minimum during AHF-D training. We utilized the Adam optimization algorithm (Jais et al., 2019) and adjusted the training image size to 416 pixels, with a batch size of

TABLE 1 Number of annotations for various fish species in Asahi Heidan Fish Dataset (AHF-D).

Fish species	Annotation numbers		
	Tain	Valid	Test
Plecoglossus altivelis (Ayu)[a]	4822	582	616
Opsariichthys platypus (Oikawa)	1258	155	166
Mugil cephalus (Bora)	2352	283	303
Lateolabrax japonicus (Suzuki)	36	3	5
Acanthopagrus schlegelii (Kurodai)	313	38	38
Tribolodon hakonensis (Ugui)	84	9	12
Silurus asotus (Namazu)	33	5	4
Hemibarbus barbus (Nigoi)	93	12	13
Cyprinus carpio (Koi)	16	9	2
Gobioidei (Haze)	82	10	12

[a]The names enclosed in parentheses represent common names for various fish species.

100 images. These configurations were determined through a process of trial and error.

2.4.3 | YOLOv5 modeling and evaluation metrics

We employed the YOLOv5 model, an AI-driven deep convolutional neural network, primarily for object detection (Li et al., 2023; Liu et al., 2023). It offers several advantages over other deep learning models. Its standout feature is its rapid inference speed. The YOLOv5 model benefits from pretraining on diverse datasets, giving it a comprehensive understanding of various object features, including diverse objects like fish. Furthermore, its lightweight design enhances computational efficiency compared to its peers (Liu et al., 2023). This efficiency leads to lower computational resource requirements, making it ideal for settings with limited computational power.

Table S4 presents the computer-related parameter settings for YOLOv5 v6.0 utilized herein, while Table S5 displays the basic library requirements for the AI-driven model. We trained the YOLOv5 model on the AHF-D dataset to evaluate its capability to differentiate Ayu from other fish species. This required analyzing test data with the trained model. It is worth noting that the dataset from the Asahi Heidan region captures unique fish features, even when the entire fish might not be visible. To gauge detection accuracy, we calculated *F*1-score values (Equation 1), precision (Equation 2), and recall (Equation 3). These metrics were based on True and False values for each fish species, as detailed in Table S6. The subscripts 1 through 10 relate to the order of species like Ayu and Oikawa, consistent with the model's classifications. For example, $\mathrm{True}_{1,1}$ signifies the proportion of fish accurately labeled as Ayu, whereas $\mathrm{False}_{1,2}$ represents those mistakenly labeled as Ayu instead of Oikawa (*Opsariichthys platypus*). During our tests, there were instances where fish species, including Oikawa and Ugui (*Tribolodon hakonensis*), were mistakenly identified as Ayu. Figure 3 showcases examples from these detection trials.

$$F1-\mathrm{score}(\mathit{Plecoglossus\ altivelis}) = \frac{2\times\mathrm{Precision}(\mathit{Plecoglossus\ altivelis})\times\mathrm{Recall}(\mathit{Plecoglossus\ altivelis})}{\mathrm{Precision}(\mathit{Plecoglossus\ altivelis})+\mathrm{Recall}(\mathit{Plecoglossus\ altivelis})}. \quad (1)$$

$$\mathrm{Precision}\,(\mathit{Plecoglossus\ altivelis}) = \frac{\mathrm{True}_{1,1}}{\mathrm{True}_{1,1}+\mathrm{False}_{1,2}+\cdots+\mathrm{False}_{1,11}}. \quad (2)$$

$$\mathrm{Recall}\,(\mathit{Plecoglossus\ altivelis}) = \frac{\mathrm{True}_{1,1}}{\mathrm{True}_{1,1}+\mathrm{False}_{2,1}+\cdots+\mathrm{False}_{11,1}}. \quad (3)$$

To further evaluate the efficacy of the YOLOv5 model trained on AHF-D in detecting Ayu, we juxtaposed results from manual visual counts against those from the YOLOv5 model using the same criteria. For a more quantitative comparison, we introduced the 'Ayu index'. This metric goes beyond merely counting individual Ayu sightings; instead, it increases by one each time an Ayu is detected. Essentially, the Ayu index becomes substantial when there is a significant presence of Ayu around the artificial spawning grounds. Its primary purpose is to represent the frequency of Ayu group sightings within specific time intervals. Ultimately, by contrasting the Ayu indices derived from manual counts with those produced by the YOLOv5 model, we can ascertain how effectively the model assists staff in monitoring Ayu aggregations during each hourly segment.

2.4.4 | Data augmentation technique employed to enhance YOLOv5-derived detection accuracy

In Figure 3b, we observed numerous instances of false detections, which included the misclassification of Ayu as Oikawa. To improve the accuracy of the fish species detection model, we employed a data augmentation approach, especially in conditions affected by both the shortage of model training dataset and the natural environment of the target site (turbidity and solar radiation). To achieve better performance, we harnessed the power of albumentations, as introduced by Buslaev et al. (2020). This feature is available as an option in YOLOv5 and offers diverse image processing techniques applied to the dataset's images based on predefined probabilities. These techniques encompass adjustments to brightness and contrast, as illustrated in Figure 4. This augmentation approach was considered effective for the Asahi Heidan region, which had changing water levels and turbidity because of its tidal characteristics.

3 | RESULTS AND DISCUSSION

3.1 | Field observations of spawning and riverbed conditions

3.1.1 | Spawning beds and egg density survey

The Ayu fish's spawning activity was first observed on 16 October 2020, coinciding with the drop in water temperature below 20°C,

(a)

(b)

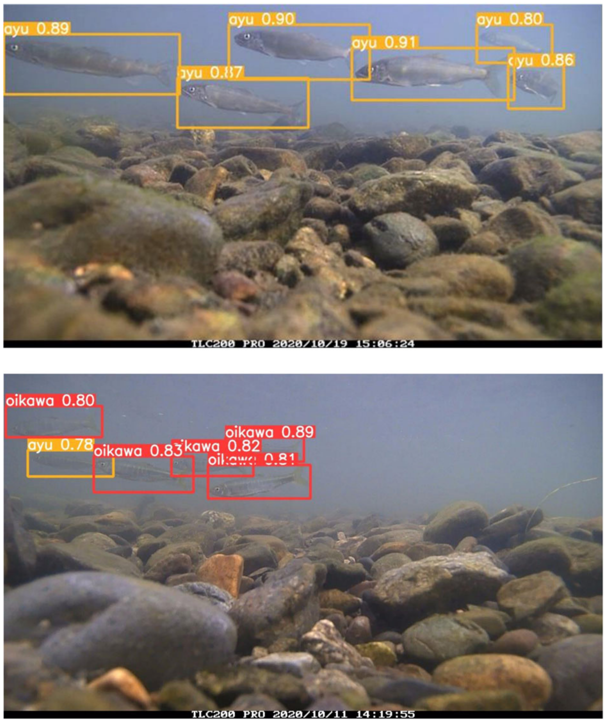

FIGURE 3 Results of detection trials conducted on a test dataset from site P1 (referenced in Figure 1d), utilizing a trained YOLOv5 model: (a) accurate identification of Ayu fishes; (b) misidentification of Oikawa as Ayu. The values in the bounding boxes represent the model's confidence scores for each detection. [Color figure can be viewed at wileyonlinelibrary.com]

shown in Figure S4. A closer examination on 21 October 2020 at four sites found eggs at two locations, P1 and P2. By 28 October, a significant increase in eggs was evident, especially at P1, highlighted in Figure S5. In this context, dots signify egg detection sites, and connecting lines indicate dense egg areas. By 13 November, 2020, the eggs had advanced to the stage of eye development. All eggs showed eye formation by 20 November. These patterns suggest that the peak of the spawning activity in these areas likely concluded around 14 November. Our observations suggest that temperature thresholds influence Ayu fish spawning, consistent with findings from studies on other fish species with temperature-dependent reproductive behaviors (Asch et al., 2019; Nagayama et al., 2023). Notably, the two spawning sites, P1 and P2, had distinct substrate properties. At P1, about 70% of the gravels were large with grain sizes over 5 cm, and many of these gravels housed multiple eggs. In contrast, P2's gravels were mainly smaller, between 1 and 3 cm, usually containing a single egg. The differences in substrate between P1 and P2, impacting egg laying and hatching, emphasize the importance of suitable habitats in fish reproduction (Daupagne et al., 2022; Duchesne et al., 2021). In addition, the study found no evidence of spawning at P3.

Egg density analysis, displayed in Figure 5a, showed that P1 and P2 both had 0 eggs/m^2 on 15 October 2020. But the following days saw a consistent increase, peaking on November 6. This trend supports our earlier observations of heightened spawning between these dates. Impressively, P1's egg density reached a peak of 48,817 eggs/m^2, more than seven times higher than P2's maximum. The notable egg density disparity between the two sites can be tied to factors like substrate quality, water flow, and temperature, known to affect spawning success in various fish species (Snickars et al., 2010; Sternecker et al., 2014). Overall, these thorough analyses suggest that, of the studied sites, P1 is the optimal habitat for Ayu spawning.

3.1.2 | Riverbed conditions survey

Figure 5b presents the grain size distribution on the riverbed at predetermined spawning sites, with a notable predominance of gravels within the 0.85–53.0 mm range. This range is delineated by a red boundary in Figure 5b and is identified as ideal for spawning, as corroborated by Ishida and Ichijo (1990) and Hyodo et al. (2014), who observed that similar gravel sizes favor various riverine species' spawning. Figure 5c delves deeper, contrasting the identified size's gravel proportions against their penetration depths. In Figure 5c, circles indicate sites with confirmed spawning activity, while crosses denote sites where spawning was not observed. Site P1, which emerged as a key egg-laying location following bed development, had 91.6% of its gravels within the optimal 0.85–53.0 mm range. Yet, it showed a relatively shallow average penetration depth of 7.4 cm, which can sometimes undermine the site's spawning potential. In contrast, site P2, post-bed development, contained 88.0% of its gravels in the preferred range and had a deeper average penetration depth of 12.4 cm. Analyzing data from Figure 5c suggests that P2 might offer a better spawning environment than P1.

An additional notable finding was the hardness of the riverbed at P1, which complicated sampling at the intended 10 cm depth. As noted by Haschenburger et al. (2007), bed hardness can introduce sampling biases, leading to the speculation that the actual proportion of suitable gravels at P1 might have been underestimated. Considering these observations and supported by the research of Nagaya et al. (2008), Hyodo et al. (2014), and Awata et al. (2020), site P2 appears to be a more favorable habitat for the spawning activities of Ayu.

3.2 | Comparison of sensor-based observations and numerical results

A thorough validation was executed on the AEM3D numerical model by meticulously comparing the collected data, including water level, salinity, and temperature, with their simulated counterparts in the designated region.

Figure 6a reveals a strong alignment between observed and simulated water levels at locations P1–P4. Regarding salinity, locations P1, P3, and P4 exhibit a compelling consistency in the data. However, Figure 6b highlights a slight deviation in accuracy for P2. This discrepancy can be attributed to the limitations of the 5 × 5 m analytical mesh, which struggled to capture the intricate features of P2—a region known for its complex topography, swift flow dynamics, and

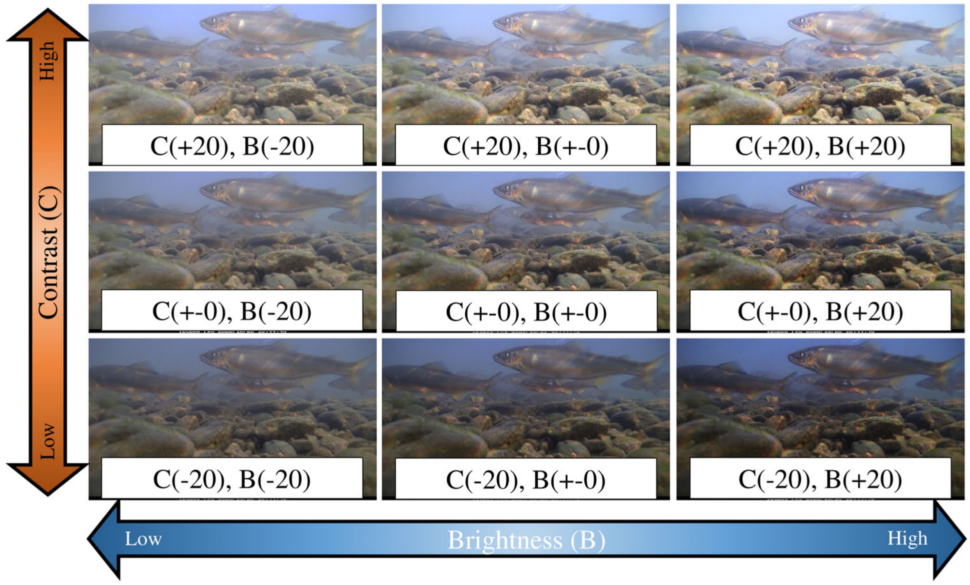

FIGURE 4 Albumentations applied in this study with specific probabilities, including brightness and contrast adjustments. [Color figure can be viewed at wileyonlinelibrary.com]

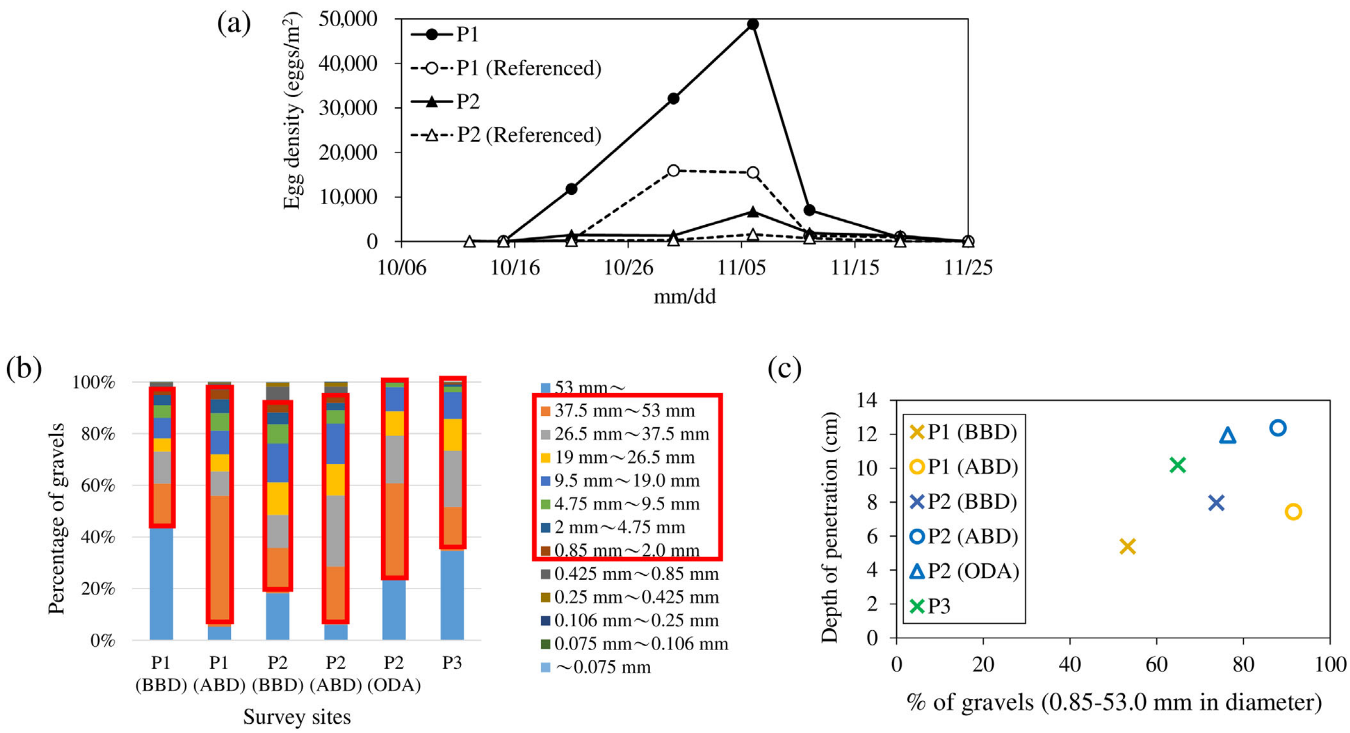

FIGURE 5 (a) Egg density at each survey site, considering the referenced areas highlighted in Figure 1d for the creation of artificial spawning grounds, (b) grain size distribution of riverbed materials at different survey sites: ABD (after bed development), BBD (before bed development), and ODA (outside the developed area); the red enclosure highlights the gravel percentage suitable for Ayu spawning (Hyodo et al., 2014; Ishida & Ichijo, 1990), and (c) percentage of gravels in the ideal size range of 0.85–53.0 mm and their penetration depth. [Color figure can be viewed at wileyonlinelibrary.com]

proximity to saltwater boundaries. Additionally, an interesting observation is the recurring intrusion of salinity at P3, with the trend closely matching both observations and numerical calculation. Consequently, drawing from this extensive comparison, it can be assertively stated that our subsequent examination of saline intrusion in the Heidan region possesses a high degree of reliability.

Regarding temperature transitions, the model accurately reflected water temperature changes at points P3 and P4 during episodes of freshwater influx and during periods of saline intrusion, as shown in Figure 6c. Yet, during the term of dropping water temperature—typified by nighttime and dawn—the modeled readings for P1 and P2 saw considerable decreases. Such disparities could stem from inherent

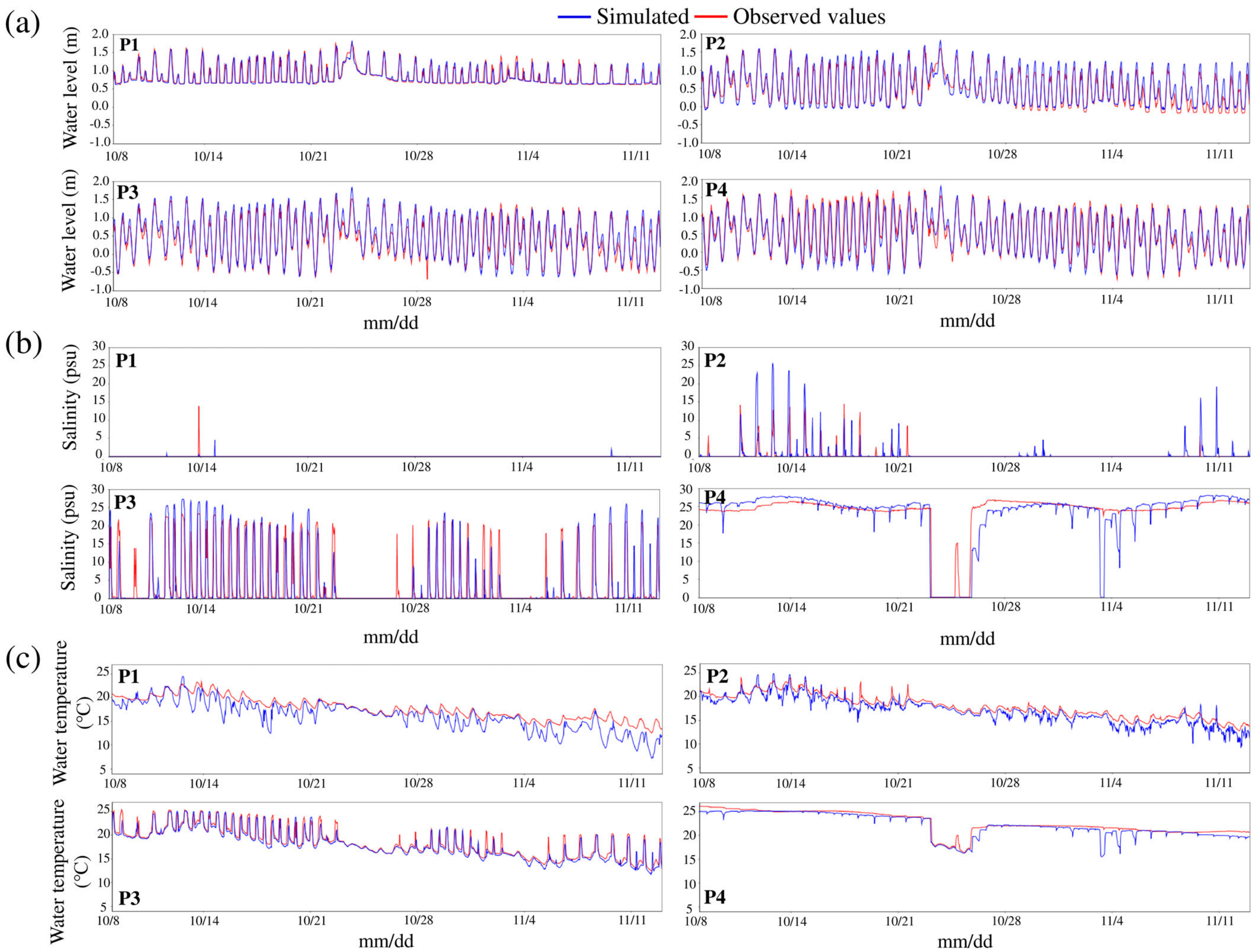

FIGURE 6 Comparison between observed values and numerical estimates for (a) water level, (b) salinity, and (c) water temperature at each target site, with salinity measurements at site P4 taken at the lower location (4.0 m below the water surface) as detailed in Supporting Information S1. [Color figure can be viewed at wileyonlinelibrary.com]

difficulties in simulating the heat exchange processes between shallow aquatic zones and the riverbed (Munz et al., 2017; Smith, 2002).

3.3 | Tidal influences on Ayu spawning

3.3.1 | Relationship between saline intrusion and Ayu spawning

Observational and computational data highlight a distinct trend of salinity levels in relation to Ayu egg density. At location P1, which exhibited a high Ayu egg density, salinity was predominantly low (Figure 6b). This pattern was mirrored at P2, albeit with lower egg density. Notably, P3, the man-made spawning ground established in 2019, consistently faced saltwater intrusion levels around 20 psu (Figure 6b).

A comprehensive spatial analysis, emphasizing peak salinity recordings in the Heidan region, confirmed that P1 was generally unaffected by saltwater influx (Figure 7a). This finding becomes particularly crucial during the primary spawning window for Ayu, identified between 15 October and 5 November 2020 from prior egg density assessments (Figure 5a). During this period, P2 experienced sporadic salinity surges of around 15 psu, particularly during the week spanning from 15 to 21 October (Figure 6b). These patterns at P2 correlate with the spatial distribution of intermittent low-salinity peaks, as depicted in Figure 7a. Intriguingly, riverbed conditions at P2, showcased in Figure 8, infer active Ayu spawning occurring in nonsaline conditions. However, actual egg densities at this site remained lower than P1, showing a significant uptick only post 30 October. This aligns with a marked decline in salinity levels (Figure 6b).

Synthesizing these results, it can be deduced that optimal Ayu spawning grounds in the evaluated region predominantly remain untouched by saline intrusion. Numerous studies have shown that salinity can impact fish egg hatching success and survival rates, particularly for species that rely on freshwater for reproduction (Alam et al., 2020; Mabidi et al., 2018). Thus, the observed patterns in our

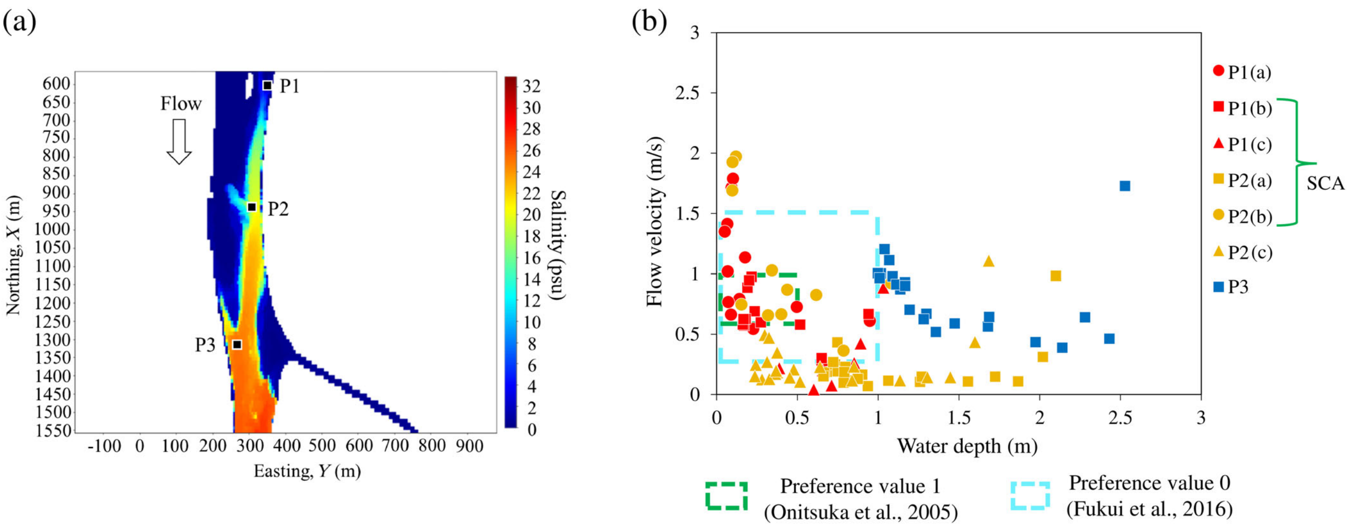

FIGURE 7 (a) Spatial distribution of peak salinity values during the period from 15 October to 5 November 2020, and (b) hourly time-averaged water depth and flow velocity estimates (recorded between 17:00 to 20:00, spanning from 15 October to 5 November 2020) at each survey site, as illustrated in Figures 1d and S5. SCA, spawning-confirmed areas. [Color figure can be viewed at wileyonlinelibrary.com]

study corroborate with the broader understanding that salinity can serve as a limiting factor for successful fish spawning (Hicks et al., 2010; Iguchi & Takeshima, 2011).

3.3.2 | Effects of water depth and flow velocity during Ayu spawning

In the zone sensitive to tidal influences, the depths and velocities of water undergo variations, including in areas located upstream from where freshwater meets saltwater. Such fluctuations could influence the spawning behavior of Ayu. In light of this, our study focused on Ayu's spawning times, typically between 17:00 and 20:00, and carried out an in-depth numerical analysis from 15 October to 5 November 2020. The results, displayed in Figure 7b, reveal the simulated water depths and flow velocities during times historically conducive to Ayu spawning. To define preferable spawning conditions, we focused on two primary preference values by utilizing the composite suitability index (CSI), a widely recognized metric in the PHABSIM model (Gard, 2009). The CSI is measured on a scale from 0 to 1, where a score of 1 indicates the most favorable environment for Ayu fish to spawn. Preference value 1 was derived from the empirical findings of Onitsuka et al. (2005), delineating certain ranges of water depth and flow velocity. In contrast, preference value 0 represents conditions where spawning has been historically observed in other river systems, as documented by Fukui et al. (2016).

As shown in Figure 5a, it is noteworthy to reiterate our result that spawning occurred at P1 and P2 but not at P3. Our research findings regarding hourly time-averaged water depth and flow velocity estimates (Figure 7b) align with the outcomes reported in prior studies by Onitsuka et al. (2005) and Fukui et al. (2016). We discovered that during the core spawning window, conditions at sites P1(a) and P1(b) predominantly matched with preference values 1 or 0. It is crucial to recognize, however, that the conditions in these earlier studies were not observed in a river-sensitive tidal area, but rather in a field with uniform flow direction. In contrast, site P1(c) presented an intriguing anomaly; despite a notable aggregation of eggs, the conditions at this site—both observed and simulated—did not align with the conditions outlined by preference value 0. This phenomenon demands further scrutiny to elucidate the factors influencing this divergence. Regarding the discrepancy with spawning at P1(c), it is important to note that the microtopography in the vicinity of P1 may not always accurately reproduce topographic data, which is a challenge associated with simulating flow conditions using a coarse computational mesh, as previously highlighted by Horritt et al. (2006), Hauer et al. (2009), and Islam, Yoshida, Nishiyama, Sakai, Adachi, et al. (2022); Islam, Yoshida, Nishiyama, Sakai, and Tsuda (2022). In comparison, the flow velocities at site P2 generally fell short of the optimal range delineated by the preference values. However, location P2(b), situated within the main channel of the river, consistently mirrored the conditions denoted by preference value 1. A significant deviation was observed at site P3, where conditions were misaligned with preference value 0. This could be attributed to the inherent deep-water characteristics of the P3 site.

In synthesizing these observations, it becomes evident that the P1 region, especially during times of increased egg density, offered favorable water depths and flow velocities for Ayu spawning. The P2(b) site, which recorded the highest spawning activity within the P2 region, further emphasized the pivotal role of optimal flow conditions in supporting Ayu's reproductive processes. The ecological ramifications of these findings are profound, accentuating the imperative to safeguard and sustain these hydraulic conditions for the conservation and restoration of the Ayu population in the Asahi River, resonating with the conservation strategies emphasized by Hata and Otake

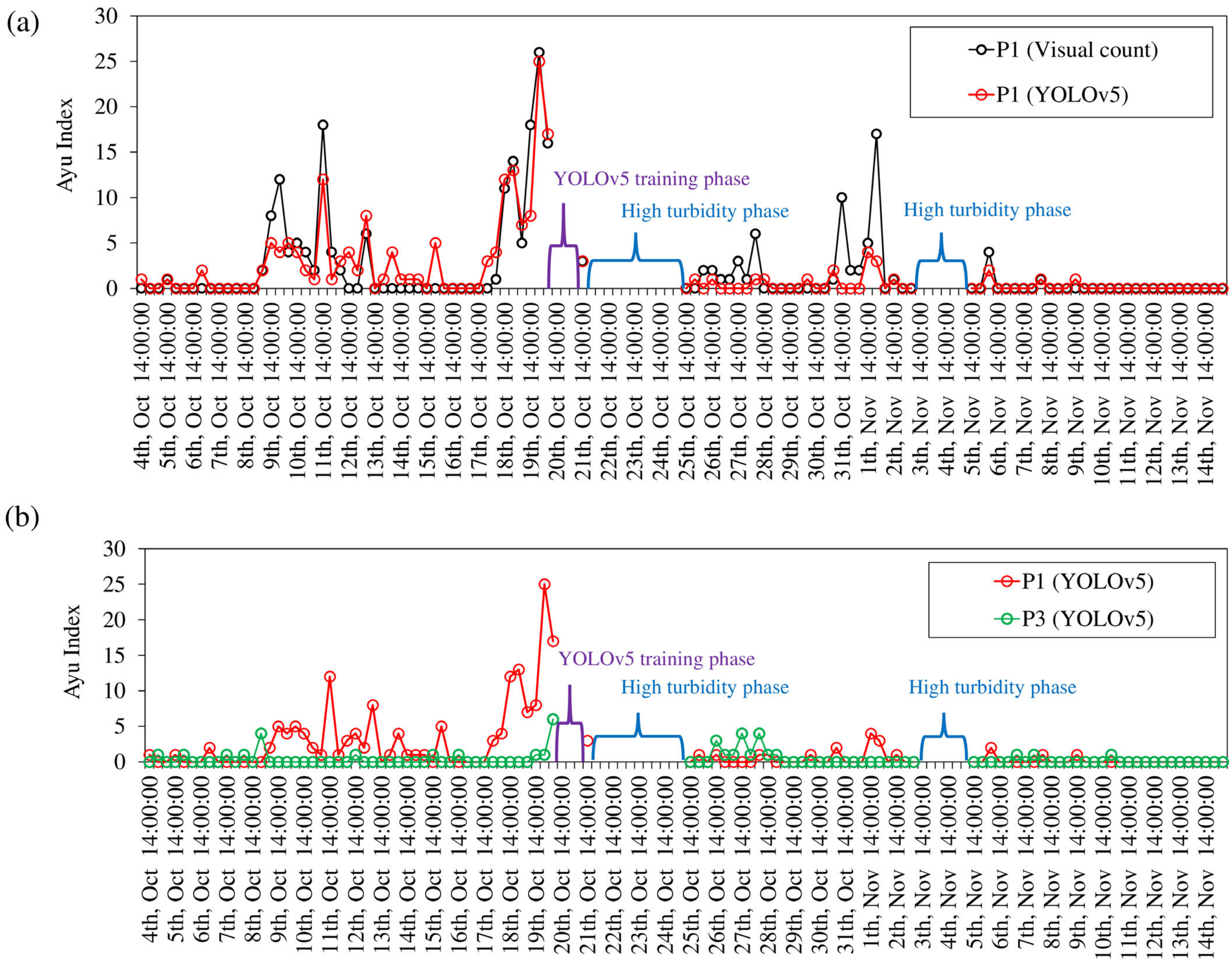

FIGURE 8 (a) Visual counts versus YOLOv5 inference results at artificial spawning ground P1, and (b) YOLOv5 inference comparisons between sites P1 and P3. [Color figure can be viewed at wileyonlinelibrary.com]

(2019) and the river management principles advocated by Osugi et al. (2007).

3.4 | AI-based detection of Ayu fish migration around artificial spawning habitats

Utilizing the YOLOv5 model, we conducted species discrimination on the AHF-D dataset, leveraging the training data outlined in Table 1. Figure S6 displays 10 distinct fish species that were the primary focus of our AI-based detection efforts. Furthermore, Table 2 provides an overview of the model's performance, specifically in terms of *F*1-scores for various fish species within the Heidan region, with particular emphasis on the Ayu species. Remarkably, the model achieved an impressive *F*1-score of 0.757 for Ayu, underscoring its proficiency in detecting this specific species. Notably, an even more remarkable *F*1-score of 0.974 was attained for the Kurodai fish (*Acanthopagrus schlegelii*). This exceptional performance can be attributed to the unique image features that distinguish Kurodai from other fish species, as presented in Figure S6. It is important to recognize that while the AI-driven model did not achieve 100% accuracy, it still demonstrates impressive capabilities in estimating the aggregation status of parent Ayu in artificial spawning grounds.

To delve deeper into Ayu fish migratory patterns around artificial spawning habitats, we conducted a comprehensive validation of the AI-driven model. This validation process involved the use of augmented AHF-D datasets, meticulously generated in accordance with the methodology outlined in Figure 4. For the validation phase, an extensive dataset comprising 38,160 images was deployed to assess the model's precision in enumerating Ayu fish. It is crucial to mention that a subset of images (3427 from the AHF-D dataset; Table S3), especially those from the training phase (Figure 8) during the entire study period (4 October and 14 November 2020), and from phases of high turbidity when the Okayama Prefectural Agriculture, Forestry, and Fisheries Research Center did not conduct official counts, were deliberately excluded from the analysis. The anticipated outcome was

TABLE 2 *F*1-scores for fish species detected by the trained YOLOv5 model on the Asahi Heidan Fish Dataset (AHF-D).

Fish species	*F*1-score
Plecoglossus altivelis (Ayu)	0.757
Opsariichthys platypus (Oikawa)	0.680
Mugil cephalus (Bora)	0.901
Lateolabrax japonicus (Suzuki)	0.500
Acanthopagrus schlegelii (Kurodai)	0.974
Tribolodon hakonensis (Ugui)	0.800
Silurus asotus (Namazu)	0.000
Hemibarbus barbus (Nigoi)	0.759
Cyprinus carpio (Koi)	0.000
Gobioidei (Haze)	0.632

that the Ayu index count would align closely with the parent Ayu count, primarily due to the model's capability to pinpoint even an individual Ayu within an image. The results, visualized in Figure 8a, predominantly corroborated with manual counts at the artificial spawning site P1. However, significant discrepancies were identified between 9 and 11 October 2020 and 25 October to 1 November 2020. These variations in results could be attributed to several factors. One potential contributing factor is the prevalence of Ayu fish images where their faces were oriented toward the underwater camera. Additionally, other factors that need consideration include instances where Ayu fish swam far away from the camera or when river turbidity was relatively high. Moreover, in human visual counting, the information used may differ from that employed by the AI method, as human counting may sometimes rely on observations from two consecutive images over time. These factors may have adversely affected the model's accuracy during those specific periods.

Furthermore, the evaluation of the YOLOv5 model involved the analysis of underwater camera images obtained from both the P3 and P1 sites. These images, spanning from 4 October to 14 November 2020, comprised a dataset of 39,923 images used for validating the model's performance. The findings, presented in Figure 8b, clearly illustrate a higher Ayu index at the P1 site compared to P3. This data reaffirms previous observations from this research, suggesting a significant migration of parent Ayu toward P1. This attraction toward P1 can likely be attributed to its amenable spawning conditions—favorable currents, lack of saline water interference, and abundant spawners. In contrast, P3 was recognized as unfriendly, with saline water intrusions and a scarcity of spawners. Predictably, Ayu concentrations were minimal at P3, resulting in a lower Ayu index when compared to the P1 spawning grounds. Broadly, these revelations resonate with the model's potential to offer insights into the behavioral patterns and spatial distribution of Ayu in the context of artificial spawning grounds in Heidan. However, it is imperative to acknowledge that certain environmental parameters (Taylor et al., 2019; Wang et al., 2021) could impact the AI-driven model's efficacy, as evident from the occasional count disparities.

4 | CONCLUSION

This research presents a comprehensive and interdisciplinary analysis of the suitability of spawning habitats for Amphidromous Ayu in the tidal zones of the Asahi River, Okayama Prefecture, Japan. The study combines field investigations, computational modeling, and AI techniques to identify key factors affecting Ayu spawning success, including water temperature, substrate composition, flow conditions, and salinity. The observations and analyses indicated a strong correlation between water temperature and Ayu spawning activity, with spawning occurring predominantly when water temperature dropped below 20°C (Takahashi & Azuma, 2006). The findings suggest that favorable spawning condition is associated with gravel compositions exceeding 60%, grain sizes ranging from 0.85 to 53.0 mm, and penetration depths of over 10 cm. Notwithstanding the construction of artificial spawning grounds, the evidence suggests a preference for natural sites, which offer greater resistance to saline intrusion and possess specific riverbed characteristics, optimal water depth, and flow velocities, resulting in improved reproductive success for Ayu fish. This study emphasizes the significance of maintaining low salinity levels, potentially devoid of saline water intrusion, to safeguard egg distribution during the crucial spawning-to-hatching phase. It also highlights the necessity of implementing appropriate flow rates to support habitat restoration tailored to species' reproductive needs. Additionally, our research demonstrates the feasibility of identifying appropriate sites for artificial spawning grounds in tidal river sections using high-resolution riverbed data obtained through the airborne lidar topo-bathymetry (ALB) technique, considering typical tidal levels and river discharge conditions. The utilization of the AEM3D model, with its detailed 5-m grid, enables accurate simulations of water levels and salinity in the target area. Furthermore, our findings on hourly time-averaged water depth and flow velocity estimates align with the preferences suggested by previous research conducted by Onitsuka et al. (2005) and Fukui et al. (2016). These insights underscore the importance of preserving favorable hydraulic conditions for Ayu conservation, in line with existing conservation strategies and river management principles. Additionally, the integration of the YOLOv5 model for fish detection represents a significant advancement in non-intrusive Ayu migration tracking, supporting conservation and habitat restoration efforts. In summary, this study advances our understanding of Ayu spawning behavior and habitat preferences while providing practical implications for conserving and restoring this species. It emphasizes the significance of habitat modification, detailed environmental monitoring, and the application of technology in managing and preserving aquatic ecosystems. These insights are invaluable for guiding future conservation strategies and actions aimed at sustaining and enhancing Ayu populations in the Asahi River and similar ecosystems.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

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